MIGUEL SUAU

Palma de Mallorca, Spain. miguel.suau@gmail.com

EDUCATION

Doctor of Philosophy in Artificial Intelligence Sep. 2018 - May 2023
Delft University of Technology

- Supervised by Frans Oliehoek and Matthijs Spaan.
- Thesis: Leveraging Factored State Representations for Enhanced Efficiency in Reinforcement Learning.

Master of Science in Mathematical Modelling and Computation Sep. 2016 - Jul. 2018
Technical University of Denmark

- Focus area: Machine learning and Scientific computing.
- Master's Thesis with Lars Kai (CogSys DTU). Reinforcement learning and multi-agent systems using Unity ML-Agents.
- GPA: 11.08/12.00.

Bachelor's degree in Energy Engineering Sep. 2010 - Feb. 2015
Technical University of Madrid *School of Industrial Engineering*

- Focus area: Energy technologies.
- Bachelor's Thesis with Carlos Veganzones (Electrical Systems ETSII): Design and construction of a small-scale wind turbine.
- Cum Laude.

SELECTED PUBLICATIONS

M. Suau, M. T. J Spaan, F. A. Oliehoek. Bad Habits: Policy Confounding and Out-of-Trajectory Generalization in RL. RLC 2024. Outstanding Paper Award on Scientific Understanding in RL.

M. Suau. Leveraging Factored State Representations for Enhanced Efficiency in Reinforcement Learning. Delft University of Technology. Ph.D. Thesis 2024.

M. Suau, J. He, M. M. Çelikok, M. T. J Spaan, F. A. Oliehoek. Distributed Influence-Augmented Local Simulators for Parallel MARL in Large Networked Systems. NeurIPS 2022.

M. Suau, J. He, M. T. J Spaan, F. A. Oliehoek. Influence-Augmented Local Simulators: A Scalable Solution for Fast Deep RL in Large Networked Systems. ICML 2022.

J. He, M. Suau, H. Baier, M. Kaisers, F. A. Oliehoek. Online Planning in POMDPs with Self-Improving Simulators. IJCAI 2022.

M. Suau, J. He, M. T. J. Spaan, F. A. Oliehoek. Speeding up Deep RL through Influence-augmented Local Simulators. AAMAS 2022.

M. Suau, J. He, E. Congeduti, J. He, R.A.N. Starre, A. Czechowski, F. A. Oliehoek. Influence-aware Memory Architectures for Deep Reinforcement Learning in POMDPs. NCAA, Springer Journal, 2022.

M. Suau, A. Agapitos, D. Lynch, D. Farrell, M. Zhou, A. Milenovic. Offline Contextual Bandits for Wireless Network Optimization Offline RL workshop, NeurIPS 2021.

N. Albers, M. Suau, F.A. Oliehoek. Using Bisimulation Metrics to Analyze and Evaluate Latent State Representations. BNAIC 2021.

J. He, M. Suau, F.A. Oliehoek. Influence-augmented Online Planning for Complex Environments. NeurIPS 2020.

M. Suau, E. Congeduti, R.A.N. Starre, A. Czechowski, F.A. Oliehoek. Influence-based Abstraction in Deep Reinforcement Learning. Workshop on Adaptive Learning Agents, AAMAS 2019.

INDUSTRY EXPERIENCE

Phaidra Oct. 2023 - Now
Lead AI Research Scientist, Remote

· Modelling and optimization of industrial systems

JP Morgan AI Research Jun. 2022 - Dec. 2022
AI Research Associate Program, London, United Kingdom

· Learning Implicit Portfolio Optimizers in Hindsight.

Huawei Research May 2021 - Oct. 2021
PhD Research Intern, Dublin, Ireland (Remote)

· Offline Contextual Bandits for Wireless Network Optimization.

Unity Technologies Jul. 2017 - Aug. 2018
Business Intelligence Analyst, Copenhagen, Denmark

· Data mining, visualization, statistical analysis, and machine learning.

NTT Data Feb. 2016 - Jul. 2016
Solutions Assistant, Madrid, Spain

· Designed and tested software to prevent money laundering for financial institutions.

SAMPOL Ingenieria y Obras Summer periods 2013 and 2014
Technician in Energy Production, Palma de Mallorca, Spain

· Energy efficiency studies in co-generation plants, biomass, fuel, and gas.

TEACHING AND SUPERVISION

Lecturer

· CSE2530, Computational Intelligence, TU Delft, 2021-2022
· CSE2530, Computational Intelligence, TU Delft, 2020-2021
· CSE2530, Computational Intelligence, TU Delft, 2019-2020

Supervision

· Honours Project: Sven Holtrop, Lucas Crijns (2020)
· Master thesis: Nele Albers (2019)
· Bachelor's Thesis: Cian Jansen (2019)
· Master thesis: Deniz Hofmeister (2018)